

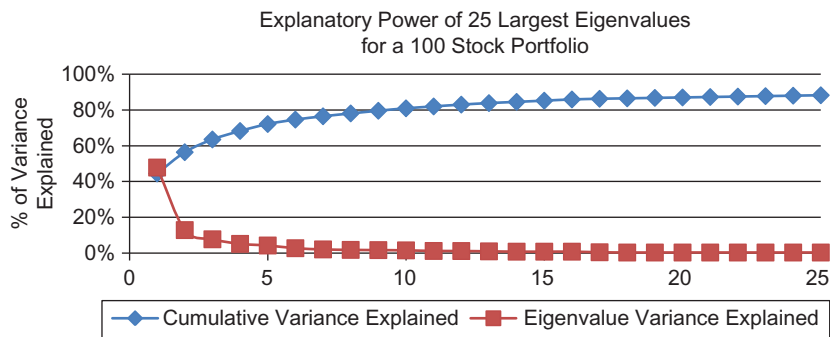
are based on fundamental and technical data. The fundamental data consists of company characteristics and balance sheet information. The technical data (also called market driven) consists of trading activity metrics such as average daily trading volume, price momentum, size, etc.

Because of the reliance on fundamental data, many authors use the term “fundamental model” instead of cross-sectional model. The rationale behind the cross-sectional model is similar to the rationale behind the macroeconomic model. Since managers and decision-makers incorporate fundamental and technical analysis into their stock selection process it is only reasonable that these factors provide insight into return and risk for those stocks. Otherwise why would they be used?

Fama and French (1992) found that three factors consisting of (1) market returns, (2) company size (market capitalization), and (3) book to market ratio have considerable explanatory power. While the exact measure of these variables remains a topic of much discussion in academia, notice that the last two factors in the Fama-French model are company specific fundamental data.

While many may find it intuitive to incorporate cross-sectional data into multi-factor models, these models have some limitations. First, data requirements are cumbersome requiring analysts to develop models using company specific data (each company has its own set of factors). Second, it is often difficult to find a consistent set of robust factors across stocks that provide strong explanatory power. Ross and Roll had difficulty determining a set of factors that provided more explanatory power than the macroeconomic models without introducing excessive multicollinearity into the data (Figure 6.9).

The cross-sectional model is derived from company specific variables, referred to as company factor loadings. The parameters are typically



■ Figure 6.9 Eigenvalue-Eigenvector Decomposition of a 100 Stock Portfolio